Weichen Wang — Curriculum Vitae

Contact Information

Department of Physics,
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Research Interests

- Early-Universe galaxy formation in extreme large-scale environments
- Galaxy and cosmic web science with current and new-generation wide-field telescopes

Education

Johns Hopkins University, Baltimore MD, United States
Department of Physics and Astronomy, 9/2016 - 12/2022
Ph. D. in Astrophysics; Thesis Advisors: Susan Kassin, Timothy Heckman

Tsinghua University, Beijing, China
Department of Physics, 8/2012 - 7/2016
B. Sc. in Physics; Thesis Advisor: Shude Mao

Research Positions

University of Milano-Bicocca, Milan MI, Italy — 2022–now
Postdoc research fellow (European Research Council funded)
Advisor: Sebastiano Cantalupo, Department of Physics
Research topics: early galaxy formation in extreme environments; high-redshift cosmic web

Johns Hopkins University, Baltimore MD, United States — 2016–2022
Graduate student; Advisors: Susan Kassin, Tim Heckman, Dept of Physics & Astronomy
Research topics: galactic winds and dust of intermediate-redshift galaxies

University of California, Santa Cruz CA, United States — 2020; 7-9/2015
Visiting student; Advisors: Sandra Faber, David Koo, Department of Astronomy
Research topics: galactic winds; star formation distribution in intermediate-redshift galaxies

Tsinghua University, Beijing, China — 2014 - 2016
Undergraduate student; Advisor: Shude Mao, Tsinghua Center for Astrophysics
Research topic: gravitational lensing models of dark matter halos

Publications as first-author or with top contributions

[5] **W. Wang**, S. Cantalupo, A. Pensabene et al. **Nature Astronomy** (March, 2025):
A Giant Disk Galaxy Two Billion Years After The Big Bang

[4] A. Pensabene, S. Cantalupo, **W. Wang** et al. **submitted to A&A** (2025):
ALMA survey of a massive node of the Cosmic Web at z∼3:
II. A dynamically cold disk galaxy in the proximity of a hyperluminous quasar

[3] **W. Wang,** S. A. Kassin, S. M. Faber, D. C. Koo et al., **ApJ**, 930, 146 (2022):
The Baltimore Oriole's Nest: Cool Winds from the Inner and Outer Parts of a Star-Forming Galaxy at $z = 1.3$

[2] **W. Wang,** S. A. Kassin, C. Pacifici et al., **ApJ**, 869, 161 (2018):
Galaxy Inclination and the IRX-β *Relation: Effects on UV Star Formation Rate Measurements at Intermediate to High Redshifts*

[1] **W. Wang,** S. M. Faber, F.-S. Liu et al., **MNRAS**, 469, 4063 (2017):
UVI colour gradients of $0.4<z<1.4$ *star-forming main-sequence galaxies in CANDELS: dust extinction and star formation profiles*

Click this ADS link for full publication list.
(26 in total with 1000+ citations; 18 in the past 3 years with 800+ citations).

Collaborations, Technical Developments, Services

Galaxy research collaborations involved:
• COSMIB (core team; VLT/JWST), • GARDEN/MEADOW (founding & core team; JWST), • SPRING (JWST), • CEERS (JWST), • HALO7D (core team until 2022; Keck).

Core team member to develop a highly efficient and **innovative observing mode** for the **James Webb Space Telescope (JWST)** NIRSpec instrument named **Slitlet Stepping**; Now incorporated into the official tech documentation, since pre Cycle 1 (2020-).

SOC/LOC member, Conference What Matter(s) Around Galaxies, 2024

Referee for The Astrophysical Journal, Astronomy and Astrophysics, etc.

Awards and Press Coverage

Research press coverage by the **European Space Agency**, **Nature Portfolio**, U. Milano-Bicocca, Caltech, Swinburne U. (reporting W. Wang et al. 2025):
"Big Wheel", a surprisingly large disk galaxy discovered in the early universe.

News coverage by Scientific American, The Independent, Newsweek (U.S.), Daily Mail (U.K.), Il Giornale (Italy) (reporting Wang+2025; along with 20+ platforms): *JWST Spots Giant Spiral Galaxy Shockingly Early in Cosmic History.*

The International Astronomical Union Travel Grant, 2019.

First-year student award, Johns Hopkins Department of Physics and Astronomy, 2016.

National Astronomical Observatory of China Scholarship, 2016.

Awards of excellence by the Tsinghua University: 清华学堂计划, 清华领军计划, 2012-2016.

Observation Programs (Selected)

JWST Cycle-4 Program (Co.I.; P.I.: B. Wang; 20 hours):
Extremely massive galaxies in the early universe? Confirming the nature of the most model-breaking object by hunting for stellar absorption features, scheduled for 2025-2026

VLT P112 Program (Co.I. & core team; P.I.: Cantalupo; 84 hours): *Connecting the dots with MUSE: the Cosmic Web in emission around a massive structure at z=3,* 2024-2025

Subaru S24A Program (Co.I.; P.I.: Liang; 2 nights): *Origin and properties of an intergalactic-scale, metal-enriched filament at z=2.3,* 2024

JWST Cycle-2 Program (Co.I. & core team; P.I.: Kassin; 68 hours): *Galaxy angular momentum alignment with filaments at $z \sim 3$: The effect of large scale structure on galaxies,* 2024

JWST Cycle-1 Program (Co.I. with major contributions, core team; P.I.: Kassin; 74 hours): *A Pathfinder for JWST Spectroscopy: Deep High Spectral Resolution Maps of Galaxies over $1<z<6$,*, 2023

JWST Cycle-1 Program (joined in 2022 with major contributions, core team; P.I.: Cantalupo; 24 hours): *Unraveling the Knots of Gaseous Cosmic Web Filaments at $z\sim3$ through H-alpha Emission Observations,* 2023

HST Cycle-30 Program (joined in 2022 with major contributions; P.I.: Cantalupo; 22 telescope orbits): *Resolving a Massive Node of the Cosmic Web at z=3,* 2023

ALMA Cycle-7 Program (P.I.) (15 hours): *Does molecular gas follow the motion of ionized gas inside typical high-redshift star-forming galaxies?* Observations not completed due to weather and the impact of COVID-19 in Chile, 2021

NASA ADAP Proposal (Co.I. with major contribution; P.I.: Kassin; $485k): *Expelling Gas from Galaxies in the Distant Universe*, 2020-2022

Summed metrics: 180+ hours on JWST as core or founding team member; 100+ hours on the Very Large Telescope (VLT) and Hubble Space Telescope (HST) as core team member.

Conferences and Talks (Selected)

Tracing Cosmic Evolution with Galaxy Clusters, Italy, 2025

(Invited talk) From Fake News to Real Clusters: The Controversial Fate of High-z Galaxy Protoclusters, Spain, 2025

Observing and Simulating Galaxy Evolution in the Era of JWST, Switzerland, 2024

(Invited talk) Galaxies and Diffuse Gas in Large-Scale Overdense Environments, Italy, 2024

(SOC/LOC member) What Matter(s) Around Galaxies 2024, Lake Como, Italy, 2024

Seminars, Tsinghua University/Peking University/NAOC, Beijing, China, 2023/2024

Astronomy Seminar, University of California, Riverside, U.S. (remote), 2021

Steward/NOIRLab Galaxy Group Lunch Talk, University of Arizona, U.S. (remote), 2021

Massively Parallel Large Area Spectroscopy from Space, Portugal (remote), 2021

Astrophysics Seminar at University of Missouri, U.S. (remote), 2020

Santa Cruz Galaxy workshop, Santa Cruz, California U.S., 2018

Mentorship

Ying Qin, Johns Hopkins undergraduate in physics major, 2021-2023 (thesis project)
Studying the Mg II emission and leaking ionizing photons from low-mass galaxies at $z \sim 1$.

M. Francesca Uboldi, Bicocca undergraduate in physics major, 2024 (thesis project):
Relations between galaxy colors and morphology based on the JWST medium-band data.

Xiaohan Wang, Visiting Ph.D student of Tsinghua University, 2025:
JWST NIRSpec studies of galaxies in the high-redshift cosmic web.

Teaching Experience

Laboratory of Data Analysis for Master Students in Astrophysics
University of Milano-Bicocca, Spring 2023

Teaching Assistant, General Physics I for Biological Science Majors (171.103)
Johns Hopkins University, Fall 2016

Teaching Assistant, General Physics Laboratory (171.111)
Johns Hopkins University, Fall 2016

Outreach Activities

Science outreach day at University of Milano-Bicocca, Italy 2024
A community event open to students from local primary/middle schools in Milan; co-designing the exhibition about astrophysical science with JWST

Member of the Astro Scholars program 2021-2022
An annual week-long program about astrophysics and computer programming for undergraduates from under-represented backgrounds; serving as a core member of the hiring & education team; monthly tag-up with the students during the rest of the year

Member of the Physics and Astronomy Graduate Students (PAGS) Outreach Team,
Johns Hopkins University, 2017-2019
Supporting visits of students from Baltimore local primary/middle schools around once per semester and teaching fundamental physics with educational demos